

Andrea Valente

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Summary

AI Computer Vision Engineer working across model development and evaluation. Builds data and training pipelines, designs experiments, and turns results into practical tools and clear technical recommendations.

Experience

Deep Vision Consulting

Modena, Italy

AI Computer Vision Engineer

Apr 2024 – Present

- **Industrial package analysis:** Became primary implementation owner of a Python analysis suite and designed and built its cloud-integrated dashboard, used by the team on new samples in operational R&D; combined signal analysis, synthetic defect data, DNN segmentation models and geometric processing.
- **3D face reconstruction:** Led the technical investigation of monocular reconstruction for a global luxury group; built synthetic-data, training and evaluation workflows that exposed accuracy and repeatability limits and helped guide product-scope decisions.
- **Point-cloud fitting:** Developed a Python package using RANSAC-based geometric fitting for smart-lighting installations, with synthetic point-cloud generation tools configurable for noise, outliers and missing faces to support evaluation.
- **Head-pose estimation:** Developed data, training, evaluation and tracking pipelines for a compact 6DoF model for browser virtual try-on; built comparison tools to support model selection with the team and client.
- **Browser inference:** Implemented project-specific ONNX Runtime/WebGPU integration and benchmarking, profiled CPU/GPU execution, and tested a pre-production path across devices.

AI Computer Vision Engineer (Intern)

Sep 2023 – Mar 2024

- Completed an MSc thesis on 3D face reconstruction for virtual try-on; compared geometric reconstruction and deformable-mesh approaches, generated synthetic training data, and trained and evaluated a face-shape estimation network, investigating synthetic-to-real limitations.

Vaia Innovations

Remote, part-time

Frontend Web Developer

Jun 2022 – Sep 2022

- Built responsive interfaces from Figma designs for article exploration and writing on a Web3 publishing platform, using TypeScript, Vue.js and Tailwind CSS.

Skills

ML and evaluation: Python, PyTorch, OpenCV, NumPy, MLflow; synthetic data, segmentation, model training and source-aware validation.

Geometric vision: 3D reconstruction, FLAME mesh fitting, camera geometry, point-cloud processing, RANSAC, SciPy optimisation, Open3D, PyTorch3D, Blender.

Inference and testing: ONNX Runtime, WebGPU, TFLite, GPU profiling, Perfetto, Playwright, cross-device testing.

Software and delivery: Git, Docker, Celery, RabbitMQ, PostgreSQL, cloud-storage integration; TypeScript, Vue.js, Tailwind CSS.

Education

MSc in Artificial Intelligence — University of Bologna

110/110 with honours

Sep 2021 – Mar 2024

BSc in Computer Science — University of Bologna

110/110 with honours

Sep 2018 – Dec 2021

Languages

Italian: native. English: professional working fluency; MSc taught in English.

IELTS Academic: overall band 7.0 (Jul 2021).